

LEO LEONG

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Currently serving National Service, available for full-time work from 2nd August 2026

SUMMARY

Cybersecurity & Digital Forensics graduate from Temasek Polytechnic with a strong foundation in software development, networking, and cybersecurity. Built full-stack software projects independently, including a café e-commerce platform and an ML-based HDB resale price predictor. Experienced in automation tooling and proof-of-concept development from an internship at Trend Micro. Seeking a software engineering role from August 2026.

SKILLS

Programming: Python, Java, JavaScript, TypeScript, PHP, HTML/CSS, SQL

Frameworks & Tools: Next.js, React, Express.js, Sequelize ORM, MQTT.js, Nodemailer, SendGrid, JWT, bcrypt, Git

Cloud & Databases: Microsoft Azure, PostgreSQL, MySQL

Cybersecurity Tools: Burp Suite, Nmap, Metasploit, Kali Linux, Wireshark

Data & ML: scikit-learn, pandas, Streamlit, Power BI

Systems: Linux (Red Hat), Nginx, Android Studio

Certificates: AWS Academy Cloud Foundations

WORK EXPERIENCE

Technical Intern

April 2023 – September 2023

Trend Micro

- Developed POC documentation for Trend Micro security products by deploying controlled VM environments, injecting test malware, and validating detection, logging, and remediation behaviors. Produced reports adopted by Sales Engineers in enterprise client presentations
- Developed a Python automation tool that extracted client data (e.g., name, email, and product details) from Excel spreadsheets and populated a standardized email template with customer-specific information for outreach campaigns. Independently designed, tested, and debugged the solution, eliminating manual one-by-one email distribution for the Sales Engineering team.

Vehicle Commander, Terrex ICV

2025 – Present

Singapore Armed Forces

- Commands a Terrex ICV independently, using the Battlefield Management System's live camera and map to direct the operator through each mission. When recon falls short and unexpected obstacles emerge on the ground, the decision falls on me alone in real time.
- Took full ownership of crew readiness, mission briefings, and tactical decision-making under operational pressure, building leadership and composure that complement technical work.

PROJECTS

EcoSaver — IoT Air Quality Monitoring Platform | *Node.js, Express, MySQL, Azure, MQTT, EJS, Nginx*

- Designed and built a full-stack air quality monitoring web platform integrating live IoT sensor data via MQTT, deployed on Microsoft Azure with a custom domain and SSL/TLS certificates
- Developed a consumer-facing website displaying real-time and historical air quality metrics (PSI, NO2, SO2, CO, O3, Temperature, Humidity) with daily and weekly data visualisation charts
- Built a separate admin management portal with dashboards for server uptime, database status, CPU monitoring, user management, sensor management, and API log tracking
- Designed and documented a RESTful API with token-based authentication, enabling third-party access to sensor data
- Configured Linux server environment including Nginx reverse proxy routing, systemd service management, and Let's Encrypt SSL certificate integration

Leo's Cafe — Full-Stack Café | *Node.js, Express, PostgreSQL, Sequelize ORM, JWT, bcrypt, SendGrid, QR Code, EJS*

- Built a production-grade F&B e-commerce platform with customer-facing ordering, a tiered loyalty programme, QR code redemptions, and table reservation management
- Implemented security-first authentication: JWT stored in httpOnly cookies, bcrypt password hashing, SendGrid email verification before account activation, and rate limiting on all auth endpoints

- Designed extensible PostgreSQL schema with Sequelize ORM; applied input validation and output sanitisation across all API routes

HDB Resale Price Predictor | *Python, pandas, scikit-learn, Random Forest, Streamlit*

- Built an end-to-end ML pipeline — data cleaning, feature engineering, and Random Forest model training on HDB resale transactions from data.gov.sg, achieving $R^2=0.843$ and MAE of S\$68,844
- Deployed an interactive Streamlit web app allowing users to input flat details and receive a future resale price estimate.

Penetration Testing Engagement | *Nmap, Burp Suite, Metasploit, Kali Linux, Wireshark*

- Conducted a simulated professional penetration testing engagement applying the Cyber Kill Chain methodology to systematically identify vulnerabilities across a target network
- Used Nmap for network reconnaissance, Burp Suite for web application testing, and Metasploit for exploitation; produced a remediation report with CVSS-scored findings and recommended countermeasures

Company Network Infrastructure Design | *Cisco Packet Tracer, VLAN, IP Subnetting*

- Designed and configured WAN connectivity between Southeast Asia offices and overseas cloud facilities for a simulated MNC, with VLAN segmentation and IP subnetting across all sites

EDUCATION

Diploma in Cybersecurity & Digital Forensics | Temasek Polytechnic

April 2021 – May 2024

- Admitted into the Director's List for AY2021/2022 for excellent academic performance
- Recipient of Edusave Certificate of Academic Achievement, 2022
- Achieved Distinctions in Network Technology and Coding & Development Project

Cambridge GCE O-Level | Pasir Ris Secondary School

2016 – 2020

- Recipient of Edusave Scholarship (2018, 2020), Edusave Certificate of Academic Achievement (2019), and Li Kaiyan Progress Award (2020)